

LUCE COUNTY AP, MI, USA

WMO: 726394

Lat: 46.311N

Lon: 85.457W

Elev: 923

StdP: 14.21

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04874

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-1.7	-11.5	3.1	-5.4	-7.9	3.7	-0.2	24.6	25.1	21.9	22.2	6.6	250	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	83.6	70.7	81.1	68.7	78.8	66.4	73.5	80.0	71.4	77.7	69.4	74.9	10.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	122.1	77.8	69.9	113.6	75.5	67.1	103.2	72.9	37.7	79.6	35.9	77.4	34.1	76.1	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.8	16.9	DB	-15.4	88.0	7.1	3.4	-20.5	90.4	-24.6	92.4	-28.6	94.3	-33.8	96.7	(4)
(5)			WB	-15.6	76.5	6.9	2.9	-20.6	78.6	-24.6	80.3	-28.4	82.0	-33.4	84.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	17.5	18.6	26.9	39.1	51.6	60.7	65.8	64.6	58.0	45.6	34.6	23.6		
	DBStd	19.28	10.50	10.70	11.06	8.87	8.20	6.80	6.15	5.66	7.79	8.13	8.63	9.64		
	HDD50	4525	1008	880	720	340	80	5	0	0	17	192	465	818		
	HDD65	8490	1473	1300	1182	776	421	164	65	77	233	604	912	1283		
	CDD50	1730	0	0	3	14	130	325	491	452	258	54	3	0		
	CDD65	220	0	0	0	0	6	34	91	64	24	1	0	0		
	CDH74	1784	0	0	1	1	87	321	796	439	136	3	0	0		
	CDH80	328	0	0	0	0	10	49	186	65	18	0	0	0		
(14) Wind	WSAvg	7.7	8.2	8.4	8.1	8.2	8.0	6.9	6.8	6.4	7.3	7.8	8.0	8.2		
(15) Precipitation	PrecAvg	31.5	2.0	1.3	1.7	2.0	2.9	3.0	2.8	3.3	3.6	3.1	2.4	2.0		
	PrecMax	41.0	5.9	3.4	5.6	5.7	7.2	6.7	7.4	8.2	8.1	6.2	5.5	3.9		
	PrecMin	23.5	0.0	0.0	0.0	0.2	0.6	0.3	0.7	0.2	1.1	0.6	0.3	0.0		
	PrecStd	4.1	1.0	0.9	1.3	1.1	1.5	1.5	1.5	1.6	1.6	1.3	1.3	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	45.9	63.0	70.5	81.2	84.3	89.6	84.4	82.1	72.6	60.7	48.5		
		MCWB	38.3	42.4	52.6	54.2	65.5	68.4	72.0	71.9	70.0	62.8	53.7	47.1		
	2%	DB	37.1	39.3	52.4	64.0	75.4	81.2	84.3	81.5	78.5	67.2	54.1	42.8		
		MCWB	35.7	36.8	46.7	50.1	61.1	68.3	71.2	69.7	67.6	61.1	50.0	40.0		
	5%	DB	35.4	36.5	45.9	58.9	72.2	78.1	81.5	79.1	74.6	63.5	50.1	38.9		
		MCWB	34.3	34.7	40.6	46.7	58.9	64.8	69.2	67.5	65.6	57.7	46.9	37.1		
	10%	DB	32.4	34.0	41.3	54.2	66.4	74.6	79.0	75.4	71.7	58.7	46.2	36.4		
		MCWB	31.6	32.3	37.0	44.5	56.0	63.3	66.7	65.4	64.6	54.4	43.8	35.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	42.1	55.2	58.0	68.7	73.7	77.1	75.1	73.1	67.0	55.1	47.2		
		MCDB	41.5	46.1	59.8	65.5	77.5	80.5	84.0	81.5	79.0	68.8	59.0	47.8		
	2%	WB	36.3	37.6	47.5	52.9	65.1	70.7	74.2	72.6	70.1	62.0	50.7	40.5		
		MCDB	37.3	39.5	51.1	60.0	72.5	78.4	81.0	78.4	74.6	65.8	53.2	42.2		
	5%	WB	33.6	34.8	41.3	49.0	61.5	67.8	71.8	70.4	67.7	58.4	46.9	37.4		
		MCDB	34.5	36.3	46.4	56.9	67.7	74.6	78.5	75.6	72.4	62.8	48.5	38.7		
	10%	WB	31.4	32.1	37.4	45.5	58.0	65.5	69.6	68.4	65.6	54.6	44.4	35.1		
		MCDB	32.4	33.6	40.5	52.7	65.1	71.1	75.5	73.7	69.6	58.0	46.5	36.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	14.7	16.2	18.0	21.4	21.5	21.1	20.0	18.7	15.3	11.8	11.6		
		MCDBR	11.7	13.8	19.4	27.3	27.9	25.0	23.9	22.5	21.1	20.2	16.6	12.5		
	5% WB	MCWBR	10.7	11.7	13.4	16.5	15.8	14.1	13.0	12.6	12.3	13.9	13.0	11.3		
		MCWBR	10.9	12.4	17.8	24.5	23.1	20.5	20.2	18.7	18.1	18.0	14.9	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.275	0.280	0.305	0.347	0.367	0.385	0.384	0.372	0.358	0.328	0.298	0.267		
		taud	2.383	2.356	2.388	2.361	2.377	2.352	2.372	2.410	2.442	2.518	2.524	2.461		
	Ebn at Noon		264	286	294	289	285	279	278	277	272	266	255	256		
		Edh at Noon	23	28	32	36	37	38	37	34	30	24	20	19		
(44) All-Sky Solar Radiation	RadAvg		344	637	1046	1421	1769	1910	1921	1626	1217	708	371	265		
	RadStd		37	68	117	159	144	130	101	106	93	82	49	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page